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QuaSAR: A Quality-Aware Selection and Proportioning Framework for Self-Supervised Deep Learning in Marine SAR Imagery

Yizhi Zhao, Jianchao Fan, *Senior Member, IEEE*

***Abstract*—Large-scale self-supervised deep learning has driven major advances in computer vision. However, pretraining with uncurated data often suffers from poor quality and diversity control, leading to longer training and weaker generalization. Moreover, downstream excessive reliance on high-quality data risks overfitting and distribution bias, while indiscriminate inclusion of lower-quality data introduces substantial noise, making effective data selection and balancing a key challenge. To address these problems, this paper proposes QuaSAR, a quality-aware selection and proportioning framework for self-supervised deep learning in marine SAR imagery. In the upstream pretraining stage, QuaSAR employs a quality-aware data selection mechanism, consisting of a multidimensional data evaluation system and an automated classification pipeline, to accelerate and stabilize representation learning. In the downstream fine-tuning stage, a principled proportioning scheme constructs fixed-size fine-tuning sets from high, medium, and low quality pools; the scheme is used to systematically analyze each quality tier's contribution to learning, and to derive recommended proportioning policies under limited annotation budgets. Extensive experiments on GF-3 and Radarsat-2 marine aquaculture SAR datasets, as well as the public MSAR dataset demonstrate that QuaSAR substantially improves self-supervised training efficiency and performance, while providing a scalable, data-centric solution for handling marine SAR data in self-supervised learning. The code supporting this work will be available at the following GitHub repository https://github.com/fjc1575/Marine-Aquaculture/tree/main/QuaSAR.**

***Index Terms*—Deep learning, data selection and proportioning, self-supervised learning, marine SAR imagery.**

I. Introduction

SELF-SUPERVISED learning (SSL) has emerged as a pivotal paradigm in computer vision [1]. Its core idea is to design pretext tasks or exploit internal data structures to learn transferable visual representations from large-scale unlabeled data. Current SSL methods can be broadly categorized into two types [2], namely contrastive learning [3], which learns discriminative representations by pulling semantically similar samples closer while pushing dissimilar ones apart, and generative learning [4], which models local structures and spatial dependencies through image reconstruction, masked prediction, or autoencoders. Compared to fully supervised learning, SSL not only alleviates the high cost of data annotation but also extracts rich semantic structures and spatial features from massive unlabeled data, enhancing model generalization and cross-domain robustness [5]. Recently, the development of deep learning architectures such as transformer has further enhanced the ability of SSL methods to integrate hierarchical features [6], capture multi-scale information [7], and learn dense representations [8]. These advances have led to significant performance improvements in tasks such as semantic segmentation, classification and object detection [9], while also providing crucial pathways for efficient utilization of large-scale unlabeled data and innovative training strategies.

Synthetic aperture radar (SAR), with its day-and-night imaging capability and robustness to cloud and fog interference, has become a key technology for ocean monitoring and resource management [10]. In applications such as marine aquaculture, these characteristics enable stable observations even under nighttime or adverse weather conditions, providing indispensable support for tasks such as target detection [11] and semantic interpretation [12]. With the rapid development of satellite remote sensing and imaging sensor technologies, the spatiotemporal resolution of SAR data has steadily improved, leading to an exponential increase in data volume [13]. However, the emergence of massive data also poses new challenges: achieving efficient information extraction and intelligent interpretation has become a critical issue. The increasing availability of large-scale high-resolution SAR data makes SSL a practically feasible approach for this domain. Nevertheless, training on all available data is not always optimal [14]. Massive uncurated data often lack quality and diversity control, prolonging training, introducing noisy samples, and ultimately constraining model generalization and practical applicability [15]. To address redundancy and noise in large-scale datasets, the natural language processing community systematizes data selection into filtering, cleaning, and mixing stages [16], utilizing models based on utility functions [17] or quality scoring [18] to optimize training data. By contrast, research in computer vision (CV) remains more disparate, primarily addressing isolated subproblems such as filtering [19], core-set selection [20], and sample weighting [21]. Notably, marine SAR imagery exhibits distinctive physical imaging characteristics, including complex surface-scattering mechanisms [22], inherent speckle noise [23], multi-polarimetric channel responses [24], and strong sea-state dependence [25],

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Y. Zhao and J. Fan are with the School of Control Science and Engineering, Dalian University of Technology, Dalian, China, 116023 (e-mail: zhaoyizhi@mail.dlut.edu.cn, fjchao@dlut.edu.cn).

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which result in data distributions fundamentally different from those of natural images. Although the above methods in CV perform well on natural images [19]–[21], they are not directly transferable to SAR image processing. As a result, merely increasing dataset size does not necessarily improve representation quality and may instead amplify domain-specific noise, thereby undermining self-supervised representation learning [26]. There is therefore an urgent need to develop efficient data selection strategies tailored to marine SAR imagery to improve pretraining representations, optimize training efficiency, and enhance the reliability and generalization of downstream semantic segmentation.

Well-curated data offer significant advantages during self-supervised pretraining [27]: their coverage of clear and canonical scenarios provides consistent and reliable learning signals, facilitating the acquisition of robust representations and yielding superior initializations for downstream tasks. However, during the downstream stage, solely relying on such data may present several limitations. These data typically cover a limited portion of the underlying distribution [28], which can encourage the model to overfit to narrow patterns and thereby constrain its generalization in complex real-world environments [29]. In contrast, data of lower quality, although often accompanied by noise and inherent uncertainty, tend to encompass more challenging and atypical scenarios. Data augmentation replicates these properties in a controlled manner by injecting perturbations, occlusions, or domain shifts, thereby providing empirical evidence that increased exposure to noisy and diverse inputs enhances robustness and out-of-distribution generalization [30]. Nonetheless, indiscriminate incorporation of lower quality data may lead the model to overfit to noise, resulting in performance degradation [31]. Existing research has primarily focused on model-centric approaches, including architecture design, regularization, and optimization strategies, while the systematic utilization and proportion of data across different quality tiers has received limited attention [32]. Striking an effective balance between the stability afforded by well-curated, low-noise data and the diversity introduced by noisier, less-refined samples remains a critical open question, requiring an understanding of how much noise can be tolerated while still learning robust representations in order to trade off data size and quality [33].

To address the aforementioned challenges, this paper proposes QuaSAR, a quality-aware selection and proportioning framework designed for SSL in marine SAR imagery. Firstly, QuaSAR employs a quality-aware data selection mechanism in the upstream pretraining stage. A multidimensional data evaluation system fits Gamma distributions between a reference image and each candidate to obtain optimal stretching parameters and compute a composite quality score, followed by an automated classification pipeline that partitions data into high, medium, and low-quality tiers to optimize dataset composition and accelerate self-supervised pretraining. Then, QuaSAR designs a principled proportioning strategy in the downstream fine-tuning stage. Training sets are constructed by sampling from different quality tiers according to specified ratios, analyzing the synergistic contributions of high, medium, and low-quality data, and ultimately providing recommended allocation proportions to ensure optimal learning performance under minimal data acquisition and annotation.

Our main contributions can be summarized as follows:

1) A quality-aware framework for SSL in marine SAR data is proposed, integrating data selection and data proportioning into a unified pipeline. This framework systematically addresses data quality issues during both pretraining and fine-tuning, improving SSL efficiency and model performance, and providing a scalable solution for efficient marine SAR data utilization.
2) A quality-aware data selection mechanism based on multidimensional quality assessment is designed. By combining Gamma distribution fitting with multiple image features, a composite quality scoring system is constructed, and an automated classification pipeline efficiently grades marine SAR data, providing high-quality data for upstream self-supervised pretraining to accelerate and stabilize representation learning, while laying the foundation for downstream data proportioning strategies.
3) A principled data proportioning strategy for downstream tasks is proposed, which systematically constructs training sets from different quality tiers and includes a method for recommending optimal data proportions. The synergistic roles of different quality data are revealed through feature-space distribution and weight visualization, guiding model optimization during fine-tuning and offering practical guidance for efficient data utilization under limited data acquisition and annotation in real-world scenarios.

The remainder of this paper is organized as follows. Section II reviews the related work. Section III provides the details of the proposed QuaSAR framework. Section IV presents the experimental results and discussions. Finally, Section V draws the conclusion.

II. Preliminary Work

A. Self-supervised Learning

SSL has garnered increasing attention in the field of marine SAR imagery, with applications ranging from object detection and classification to semantic segmentation. Li *et al*. [34] proposed a self-supervised network with self-distillation for oriented SAR ship detection. Li *et al*. [35] proposed SARATR-X, the first foundation model for SAR ATR, using self-supervised learning to enable label-efficient SAR target detection and classification. Fan *et al*. [36] introduced a self-supervised transformer with feature fusion for semantic segmentation in marine aquaculture monitoring. While these methods have demonstrated the efficacy of SSL in marine SAR applications, they predominantly tend to treat training data as homogeneous, largely overlooking the critical impact of variations in image quality caused by speckle noise, polarimetric responses, and sea-state conditions. In contrast, rather than modifying SSL architectures, we focus on the data aspect by introducing a quality-aware framework that systematically selects and proportions data, enhancing pretraining efficiency and downstream fine-tuning performance in marine SAR imagery.

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B. Data Selection

Data selection has been extensively studied as a means of improving data efficiency and model performance in machine learning. The overarching goal is to identify and leverage the most valuable samples within a dataset, thereby reducing training costs, mitigating noise, and enhancing generalization. Existing research in CV processing typically falls into three primary categories: filtering, core-set selection, and sample weighting. Filtering aims to remove redundant, low-quality, or harmful samples from the dataset. Oquab *et al*. [37] constructed a large curated dataset by retrieving samples from uncurated web data, removing duplicates and overlaps. Core-set selection aims to select a small, representative subset that approximates the full dataset. Xia *et al*. [38] introduced Moderate Coreset, which considers both representativeness and learning difficulty for efficient data selection. Zheng *et al*. [39] proposed coverage-centric coreset selection to preserve feature-space coverage under high pruning rates. Zheng *et al*. [40] further developed a label-free coreset selection strategy using proxy training dynamics. Sample weighting assigns each training sample a weight based on factors like learning difficulty, uncertainty, or estimated influence on model performance, enabling the model to focus on more informative examples during training. Shu *et al*. [41] proposed CMW-Net, a class-aware sample weighting method that improves robustness to label noise and class imbalance. While these techniques have demonstrated clear benefits in natural images, they are not directly transferable to SAR imagery because of its domain-specific characteristics.

In the SAR domain, prior work has investigated image-quality assessment. Zhang *et al*. [42] proposed an entropy-based re-sampling method for SAR target detection that evaluates image quality via variance-weighted entropy to guide sampling and improve performance. Huang *et al*. [43] introduced a SAR quality assessment combining objective metrics and subjective labels, optimized with IHHO-XGBoost, to pre-filter low-quality images and enhance detection. These studies have mainly focused on evaluating SAR image quality to filter out low-quality data, but they lack a systematic, multi-level stratification of image quality for tiered data utilization. In contrast, we introduce a quality-aware data selection framework grounded in multidimensional quality evaluation. By integrating Gamma-distribution fitting with various image features, a composite quality scoring system is established, and an automated classification workflow is applied to efficiently categorize large-scale marine SAR data into distinct quality tiers. This approach provides reliable high-quality samples for upstream self-supervised pretraining, promoting faster convergence and more stable representation learning.

C. Data Proportioning

Data proportioning has been studied in various machine learning scenarios, such as class balancing [44], labeled versus unlabeled data ratios in semi-supervised learning [45], and the distribution of training data across clients in federated learning [46]. However, most existing works focus on proportioning at the category or annotation level, with few studies considering stratified proportioning based on intrinsic data quality. In marine SAR imagery tasks, image quality is often affected by speckle noise, scattering mechanisms, and sea-state conditions. Even within the same semantic category, significant quality variations exist among samples, which can substantially impact model generalization and robustness. This work therefore introduces a novel research perspective: quality-aware data proportioning for downstream fine-tuning. In the QuaSAR framework, marine SAR imagery is partitioned into high, medium, and low-quality tiers, and a principled data proportioning strategy is designed to combine samples from different tiers during downstream fine-tuning. This approach balances stability, diversity, and robustness. Unlike conventional "filter-and-discard" data handling methods, our strategy preserves the potential value of low-quality samples, systematically analyzes the marginal contributions of each quality tier to model performance, and provides generalizable recommendations for proportioning. To the best of our knowledge, this study is the first to explicitly formulate and systematically address the problem of quality-aware data proportioning in marine SAR imagery.

III. Proposed Method

The overall framework of the proposed QuaSAR, consisting of a quality-aware data selection mechanism module and a principled data proportioning strategy module, is illustrated in Fig. 1. In the quality-aware data selection mechanism (QADSM) module, the gray-level histogram of each image is fitted against a reference image using Gamma distribution. Image metrics are then computed on the fitted images, followed by correlation analysis to eliminate redundancy. A composite quality score is derived from the retained metrics and Gamma fitting errors, and clustering is applied to stratify the dataset into high, medium, and low-quality samples, which are subsequently used to train an automated classification pipeline. High-quality data are utilized for self-supervised pretraining to ensure accelerated and stable representation learning. In the principled data proportioning strategy (PDPS) module, the training set size remains fixed while varying ratios of high, medium, and low-quality samples are combined to reveal roles of different quality data. Model performance heatmaps and marginal effect analyses are then employed to provide recommended allocation proportions, ensuring optimal learning performance under minimal data acquisition and annotation.

A. Quality-aware Data Selection Mechanism Module

The quality-aware data selection mechanism module is designed to perform quality grading of marine SAR data, providing high-quality samples for upstream self-supervised pre-training to accelerate and stabilize representation learning, while laying the foundation for downstream data proportioning strategies. As illustrated in Fig. 1, it mainly consists of gamma distribution fitting, a composite quality scoring system, and an automated classification pipeline.

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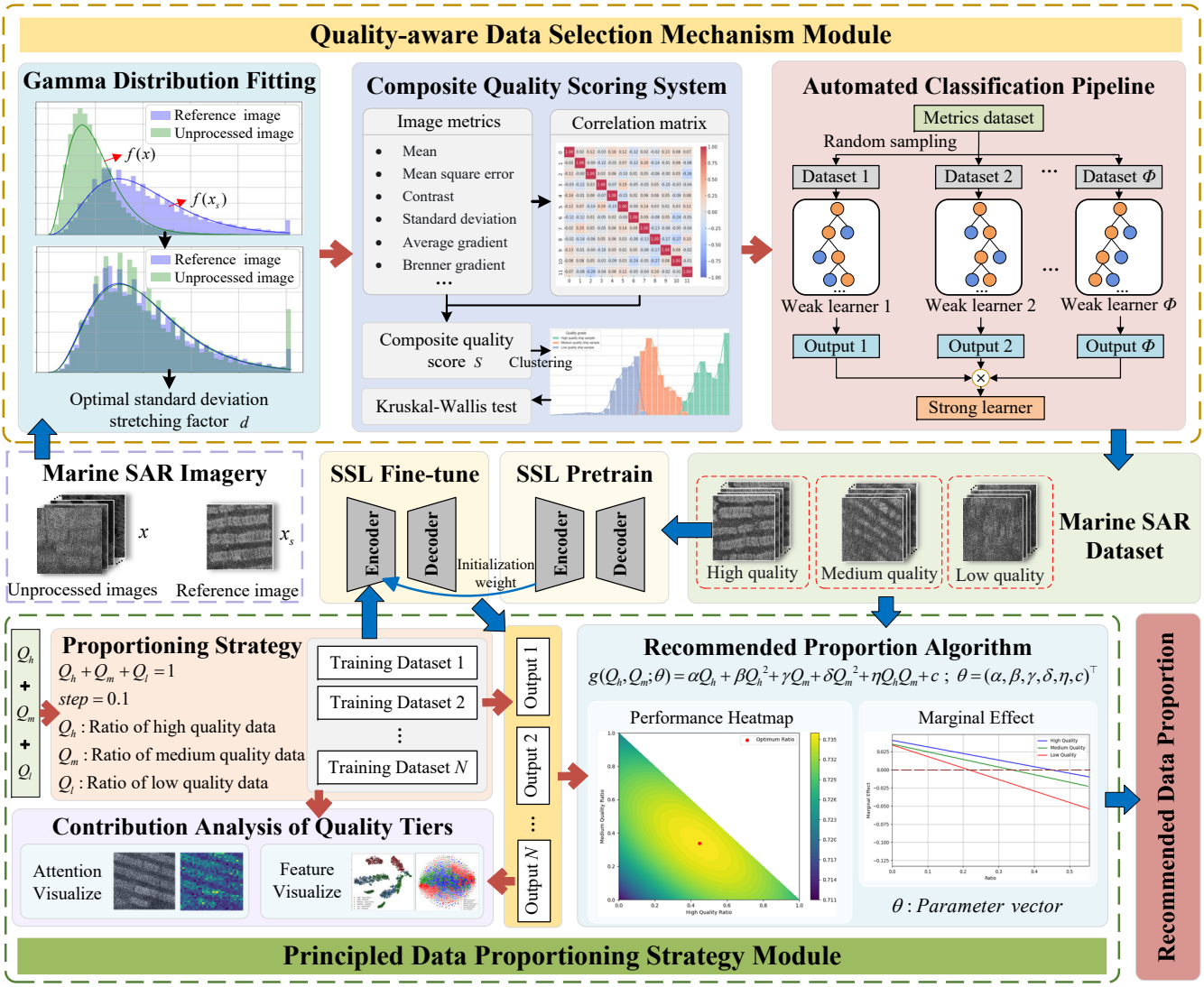


Fig. 1. Overview of the proposed QuaSAR framework. It consists of a quality-aware data selection mechanism module and a principled data proportioning strategy module. First, marine SAR images are fed into the quality-aware data selection mechanism module to construct high, medium, and low-quality datasets, and part of the high-quality data is used for self-supervised pretraining. The weights trained there are used as initialization for downstream tasks. Using the principled data proportioning strategy module, the high, medium, and low-quality data are proportioned and trained, and a recommended proportion is ultimately obtained.

1) Gamma Distribution Fitting: Due to differences in data sources and acquisition times, the grayscale means of SAR images vary significantly, resulting in dispersed distributions of subsequent image metrics and hindering effective comparison. To address this, a marine SAR image with the best visual quality—evaluated based on contrast, texture sharpness, and target discernibility—is selected as the reference image x_s. Gamma distributions are then fitted to the grayscale histograms of both the reference image and the image to be processed x_{row}. The optimal standard deviation stretching factor d is determined by minimizing the squared difference between their gamma distribution fits. The stretching formula is

$$x = \frac{x_{row} - (ave - d \cdot std)}{2d \cdot std} \cdot 255 \tag{1}$$

where x denotes the SAR image after d times standard deviation stretching, ave and std represent the mean and standard deviation of x_{row}, respectively. The optimal stretching factor d is obtained by solving the following optimization problem:

$$\min_{d} \left(f(x) - f(x_s)\right)^2 \tag{2}$$

where $f(x)$ is the probability density function of the gamma distribution:

$$f(x) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx} \tag{3}$$

where a is the shape parameter, b is the rate parameter, and $\Gamma(a) = \int_0^\infty x_{row}^{a-1} e^{-x_{row}} dx_{row}$. Here, the Gamma distribution is used as a lightweight normalization tool rather than a universal sea-clutter model. For heterogeneous sea clutter, poor Gamma fitting is reflected by the fitting error and incorporated into the composite quality score.

2) Composite Quality Scoring System: The image metrics employed for quality assessment include mean ($Mean$), mean

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square error (MSE), standard deviation (STD), contrast (C), equivalent number of looks (ENL), radiometric resolution (RL), gray-level percentile ratio (GL), average gradient (G), Brenner gradient (B), Vollath function (V), signal-to-noise ratio (SNR), and information entropy (IE). After fitting the gamma distribution, the corresponding image metrics are computed and subsequently normalized to eliminate dimensional discrepancies, thereby ensuring comparability across different metrics. Following this, a correlation analysis is performed by calculating the Spearman correlation coefficient matrix among the metrics. Redundant metrics with absolute correlation coefficients greater than 0.9 are removed, leaving a total of m remaining image feature metrics.

Considering the remaining image feature metrics and the gamma distribution fitting errors computed with respect to the reference image, a composite quality score S is defined to evaluate the quality of marine SAR imagery.

Definition 1 (Composite Quality Score)**.** Let M_i denote the i-th standardized image feature metric from a reduced set of m non-redundant metrics. Principal component analysis (PCA) is performed on the standardized feature data, and the first principal component loading vector is adopted as the weight vector ω_i. The composite quality score S is then defined as the normalized weighted fusion of these metrics:

$$S = \mathcal{N}\left(\sum_{i=1}^{m} \omega_i \cdot M_i\right) \tag{4}$$

where $\mathcal{N}(\cdot)$ denotes a normalization function mapping the weighted sum to a standardized range.

Definition 2 (Quality Tiers)**.** Based on the composite quality score S, quality tiers are determined through the following classification rule:

$$\text{Quality Tier} = \begin{cases} \text{Low-quality sample}, & S < \lambda_1 \\ \text{Medium-quality sample}, & \lambda_1 \le S < \lambda_2 \\ \text{High-quality sample}, & S \ge \lambda_2 \end{cases} \tag{5}$$

where λ_1 and λ_2 represent predetermined classification thresholds, which are calculated as

$$\begin{aligned} &\arg\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} \sum_{S_i \in C_k} (S_i - \mu_k)^2 \\ &\lambda_j = \frac{\mu_j + \mu_{j+1}}{2}, \quad j = 1, 2 \end{aligned} \tag{6}$$

where S_i denotes the composite score of the i-th sample, with $S_i \in [0, 1]$, $i = 1, 2, \ldots, n$, and n is the total number of samples. C_k represents the k-th cluster, K is the total number of clusters, and $\mu_k = \frac{1}{|C_k|} \sum_{S_i \in C_k} S_i$ is the mean of the k-th cluster. Finally, the Kruskal–Wallis (K-W) test is conducted to assess the significant differences among high, medium, and low-quality samples for each metric.

3) Automated Classification Pipeline: Due to variations in satellite data from different sources for the same target, fixed thresholds may not generalize. Therefore, an automated classification pipeline is trained using the output of the composite quality scoring system. The pipeline employs Bagging with decision trees as base learners, uses the image-metric matrix as input R, and treats the quality tier division results as training labels D. First, the metric matrix R is randomly sampled with replacement ϕ times to generate ϕ subsets $R_1, R_2, \ldots, R_\phi$, each the same size as the original training set. A weak learner g_i is trained on each subset R_i ($i = 1, 2, \ldots, \phi$), and the final prediction is given by $G(R) = \arg\max \sum_{i=1}^{\phi} \mathbb{I}(g_i(R_i) = D_i)$. The classification results of a large number of marine SAR images are $X_{class} = \{x_h, x_m, x_l\}$, where x_h, x_m, and x_l represent high, medium, and low-quality samples, respectively.

B. Principled Data Proportioning Strategy Module

The principled data proportioning strategy module aims to determine the role of training data with different quality tiers in the learning process and provides a recommended method for setting their proportions, ensuring optimal learning performance with minimal data acquisition and labeling. The module consists of three components: the proportion strategy, contribution analysis of quality tiers, and the recommended proportion algorithm.

1) Proportioning Strategy: To systematically analyze the impact of data quality on model performance, a proportioning strategy tailored for downstream fine-tuning stage is designed. Specifically, the training dataset is divided into three subsets—high-quality x_h, medium-quality x_m, and low-quality x_l. Excluding the portion of high-quality data used for pretraining, and under the constraint of a fixed total sample size, multiple training set configurations are constructed by randomly sampling different combinations of (Q_h, Q_m, Q_l) from the remaining high, medium, and low-quality subsets, subject to the following constraints: $0 \le Q_h \le 1, 0 \le Q_m \le 1, 0 \le Q_l = 1 - Q_h - Q_m \le 1$. Here, Q_h, Q_m, and Q_l denote the proportions of high, medium, and low-quality data, respectively. For each proportion scheme, performance is evaluated using task-specific metrics. The resulting experimental outcomes form the foundation for subsequent contribution analysis of quality tiers and for the recommendation of optimal data proportioning.

2) Contribution Analysis of Quality Tiers: Starting from the network initialized with SSL-pretrained weights, the downstream network is fine-tuned in three sequential stages: it is first fine-tuned using medium-quality data only; high-quality data are then incorporated to continue fine-tuning; and finally, low-quality data are introduced into the training process. During this process, the weight distributions corresponding to the three data types are visualized to reveal their relative contributions in optimization. Furthermore, based on the weights obtained through this procedure, the feature spaces of high, medium, and low-quality data are also visualized by t-SNE, providing an intuitive demonstration of the differences and complementarities in feature representations across different quality tiers.

3) Recommended Proportion Algorithm: Based on the results of proportioning experiments, a parameterized fitting model of performance index versus data quality proportions is established. Through constrained optimization and analysis of marginal effects, a reasonable range of optimal data pro-

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portions is further determined. The model adopts a quadratic regression form with interaction terms, expressed as

$$\begin{aligned} g(Q_h, Q_m; \theta) &= \alpha Q_h + \beta Q_h^2 + \gamma Q_m + \delta Q_m^2 + \eta Q_h Q_m + c \\ \theta &= (\alpha, \beta, \gamma, \delta, \eta, c)^{\mathsf{T}} \end{aligned} \tag{7}$$

where $g(Q_h, Q_m; \theta)$ is a parameterized fitting model of performance index with respect to data quality proportions, θ is parameter vector. Since Q_l can be determined from Q_h and Q_m, it is not included in the above (7). The design matrix $\boldsymbol{X}$ and observation vector $\boldsymbol{y}$ are defined as

$$\begin{aligned} \boldsymbol{X} &= \begin{bmatrix} Q_{h,1} & Q_{h,1}^2 & Q_{m,1} & Q_{m,1}^2 & Q_{h,1}Q_{m,1} & 1 \\ Q_{h,2} & Q_{h,2}^2 & Q_{m,2} & Q_{m,2}^2 & Q_{h,2}Q_{m,2} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \\ Q_{h,N} & Q_{h,N}^2 & Q_{m,N} & Q_{m,N}^2 & Q_{h,N}Q_{m,N} & 1 \end{bmatrix} \\ \boldsymbol{y} &= [y_1, y_2, \ldots, y_N]^{\top} \end{aligned} \tag{8}$$

where y_i denotes the performance index of the i-th proportioning experiment, and N is the total number of proportion combinations. The residual vector is $\boldsymbol{r}(\theta) = \boldsymbol{y} - \boldsymbol{X}\theta$, and the overall error function is $S(\theta) = \boldsymbol{r}(\theta)^{\top}\boldsymbol{r}(\theta) = \sum_{i=1}^{N}(y_i - g(Q_{h,i}, Q_{m,i}; \theta))^2$. The parameter estimates are obtained by solving the equations where the gradient of the error function is set to zero, $\partial \mathrm{S}/\partial\theta = -2\boldsymbol{X}^{\top}(\boldsymbol{y} - \boldsymbol{X}\theta) = 0 \Rightarrow \hat{\theta} = (\boldsymbol{X}^{\top}\boldsymbol{X})^{-1}\boldsymbol{X}^{\top}\boldsymbol{y}$. Based on the fitted parameters $\hat{\theta}$, the predicted values are $\hat{y}_i = g(Q_{h,i}, Q_{m,i}; \hat{\theta})$. The recommended proportions (Q_h^*, Q_m^*) are obtained by solving the corresponding optimization problem and may vary with changes in the dataset.

$$\begin{aligned} (Q_h^*, Q_m^*) &= \arg\max_{Q_h, Q_m} g(Q_h, Q_m; \hat{\theta}) \\ Q_l^* &= 1 - Q_h^* - Q_m^* \end{aligned} \tag{9}$$

The marginal effect is defined as the partial derivative of the performance index with respect to each proportion, expressed as

$$\begin{aligned} ME_h &= \frac{\partial g}{\partial Q_h} = \alpha + 2\beta Q_h + \eta Q_m \\ ME_m &= \frac{\partial g}{\partial Q_m} = \gamma + 2\delta Q_m + \eta Q_h \\ ME_l &= \frac{\partial g}{\partial Q_l} = -(ME_h + ME_m) \end{aligned} \tag{10}$$

where ME_h, ME_m and ME_l are the marginal effects of high, medium, and low-quality data proportions on the performance index, respectively. Intuitively, the marginal effect characterizes the performance gain (or loss) brought by a unit increase in the proportion of a specific data tier, while keeping the total size fixed. The obtained recommended proportions and the marginal effects of each quality tier can guide the design of training sets and optimize data utilization for improved model performance. The procedure of QuaSAR, integrating the quality-aware data selection mechanism and the principled data proportioning strategy described above, is summarized in **Algorithm** 1.

Algorithm 1 QuaSAR: Quality-aware Selection and Proportioning Framework

Require: Unlabeled marine SAR image set x_{row}; reference image x_s; small labeled fine-tuning set
Ensure: Fine-tuned downstream model and optimal proportion (Q_h^*, Q_m^*, Q_l^*)
Stage 1: Quality-aware Data Selection (QADSM)
for each image $x_i \in x_{row}$ **do**
  Normalize x_i via Gamma-distribution fitting with x_s using (1)–(3)
  Extract image-quality metrics
end for
Perform correlation analysis and retain non-redundant metrics
Compute PCA weights and obtain composite quality score S_i using (4)
Cluster $\{S_i\}$ to derive quality thresholds and assign quality tiers using (5)–(6)
Train a quality classifier based on the clustering results and partition x_{row} into x_h, x_m, and x_l
Stage 2: SSL Pretraining
Pretrain encoder using only x_h
Obtain pretrained weights $W_{pretrain}$
Stage 3: Principled Data Proportioning (PDPS)
for all proportion combinations (Q_h, Q_m, Q_l) **do**
  Construct fine-tuning set according to (Q_h, Q_m, Q_l)
  Initialize downstream network with $W_{pretrain}$
  Fine-tune and evaluate performance
end for
Fit quadratic regression model $g(Q_h, Q_m; \theta)$ using (7)–(8)
Compute optimal proportion (Q_h^*, Q_m^*, Q_l^*) via (9)
Construct final training set using (Q_h^*, Q_m^*, Q_l^*)
Fine-tune downstream model and obtain final weights
return Fine-tuned model, (Q_h^*, Q_m^*, Q_l^*)

IV. Experiments and Discussion

To validate the effectiveness of the proposed QuaSAR method under the SSL framework, it is applied to two downstream tasks: semantic segmentation and object detection. Experiments are conducted on SAR marine aquaculture segmentation datasets from two different sensors as well as on a public dataset for object detection. In the data selection process, the correlation analysis among metrics, the results of data quality division, the performance of the automated classification pipeline, and the QADSM module performance are systematically analyzed. Finally, within the principled data proportioning module, the results of data proportioning experiments, the contributions of different quality data and the recommended proportions are analyzed in detail.

A. Datasets and Implementation

1) Semantic Segmentation Datasets: Floating raft aquaculture, as a typical form of offshore aquaculture, involves constructing floating raft frames in shallow sea areas and suspending cultured organisms, primarily for the cultivation of economically valuable species such as algae and shellfish. In SAR imaging, the double-bounce scattering characteristics of floating rafts are significantly stronger than those of the sea surface, resulting in high contrast between rafts and

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the surrounding seawater background, which facilitates target identification and extraction. The dataset used in this study is derived from marine SAR images acquired by the GF-3 and Radarsat-2 satellites, both in HH polarization mode. The GF-3 satellite provides a spatial resolution of 3 m, while the Radarsat-2 satellite offers a spatial resolution of 8 m, with coverage over Changhai County, Liaoning Province, China. The raw images underwent preprocessing steps including radiometric calibration, geometric correction, and noise reduction, followed by the application of a land mask. Ultimately, a total of 12,135 GF-3 SAR images and 3,820 Radarsat-2 SAR images are obtained, each with a size of 512 × 512 pixels.

2) Object Detection Datasets: The public dataset used in this study is the large-scale multi-class SAR image target detection dataset (MSAR-1.0) [47], which consists of 28,449 detection slice samples derived from HISEA-1 and GF-3 satellites. The MSAR-1.0 dataset includes four polarization modes: HH, HV, VH, and VV. It covers diverse scenes such as airports, ports, coastal areas, islands, open seas, and urban regions, and contains four categories of targets: airplanes, oil tanks, bridges, and ships. Since the proposed QuaSAR method is specifically designed for marine SAR imagery, slice samples containing land-based targets are removed, and only ship slice samples are retained, resulting in a total of 18,292 image slices, each with a size of 256 × 256 pixels.

3) Implementation Details: An upstream–downstream architecture is constructed based on a vision transformer (ViT) backbone. The upstream network utilizes self-supervised pretraining with DINO[1], while the downstream network employs UPerNet for semantic segmentation and Cascade R-CNN with a ViT backbone for object detection. All experiments are conducted in PyTorch 1.12.1, using an Intel Xeon Silver 4210R with a clock speed of 2.4 GHz and an Nvidia GeForce RTX 3080.

4) Performance Measures: For semantic segmentation, mean intersection over union ($mIoU$), overall accuracy (OA), $Kappa$ coefficient, F_1-score, precision (P), and recall (R) are used, with $mIoU$ serving as the primary metric. For object detection, average precision (AP) metrics are adopted, including AP_{50}, AP_{75}, AP_S, AP_M, and AP_L, where AP_{50} and AP_{75} correspond to IoU thresholds of 0.50 and 0.75, and AP_S, AP_M, and AP_L evaluate targets at different scales. These metrics enable a comprehensive evaluation of QuaSAR across both downstream tasks.

B. Quality-aware Data Selection

1) Correlation Analysis of Image Metrics: A correlation analysis is conducted on various image quality metrics of the marine SAR aquaculture dataset after Gamma distribution fitting. Spearman's rank correlation coefficient is used to evaluate the nonlinear dependencies among the metrics. The resulting correlation coefficient matrix is shown in Fig. 2. To avoid redundancy caused by high correlations among the image metrics, further screening is performed. Metrics with an absolute correlation coefficient greater than 0.9 are removed, and finally, 5 representative metrics are retained: $Mean$, MSE, C, G, and IE.

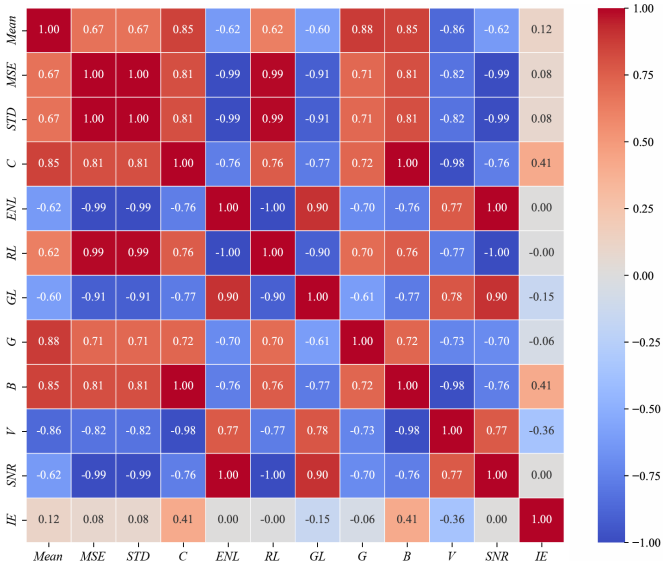


Fig. 2. Spearman correlation coefficient matrix of image metrics in the marine SAR aquaculture semantic segmentation dataset.

2) Data Quality Division: Since seawater samples are not removed during dataset preprocessing, the GF-3 data is divided into four tiers: high-quality aquaculture samples, medium-quality aquaculture samples, low-quality aquaculture samples, and seawater samples. Subsequently, a composite scoring metric for the dataset is calculated by combining the retained image metrics with the Gamma fitting error using (4). Based on this, the thresholds for classifying samples into different tiers are determined using (6). The clustering results of the composite score and boxplot of sample distribution across clusters are shown in Fig. 3.

As shown in Fig. 3(b), some outliers occur between seawater and low-quality aquaculture samples. This is mainly due to some seawater samples are affected by natural factors such as wind speed and waves, leading to local brightness variations or structural ripples in the images. These features are somewhat similar to those of low-quality aquaculture in certain dimensions, causing the clustering model to produce blurred category boundaries. Although such seawater samples overlap with low-quality aquaculture in the feature space, their composite scores are still generally lower than those of low-quality aquaculture samples. In addition, distance-based clustering is sensitive to class imbalance; the large proportion of seawater samples may pull the cluster center toward dense seawater regions, shifting the low-quality aquaculture center and reducing the separation threshold. To alleviate this issue, a center adjustment strategy $C_{adjusted} = C_{original} \times (1 - h) + C_{mid} \times h$ is introduced to correct the boundary between seawater and low-quality aquaculture categories. Here, $C_{original}$ denotes the original cluster center, C_{mid} denotes the mean of cluster centers excluding seawater and low-quality aquaculture, $C_{adjusted}$ denotes the adjusted cluster center, and h represents the adjustment coefficient. In this study, setting h = 0.25 yields a better separation between seawater and low-quality aquaculture samples. The

[1]https://github.com/facebookresearch/dino

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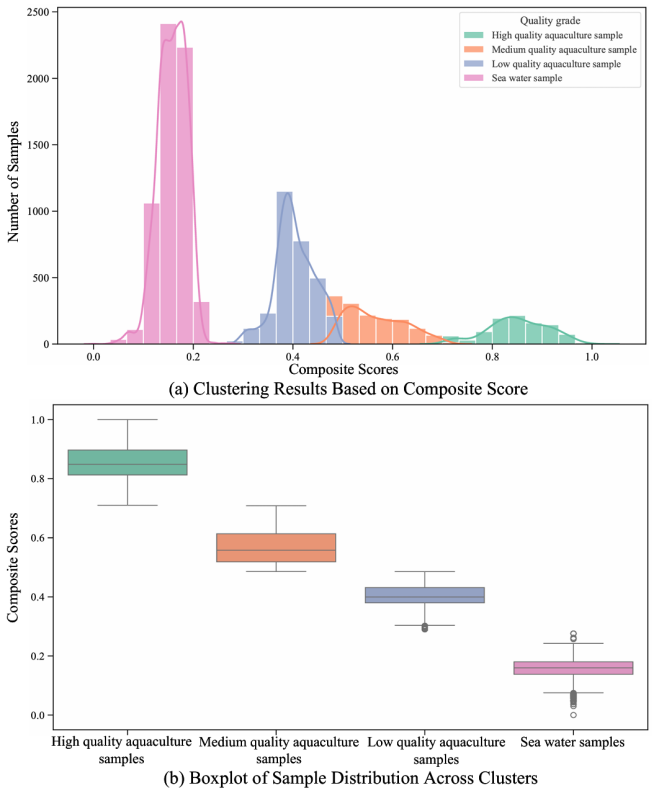


Fig. 3. Clustering results of the composite score and boxplot of sample distribution across clusters in GF-3 marine SAR aquaculture dataset.

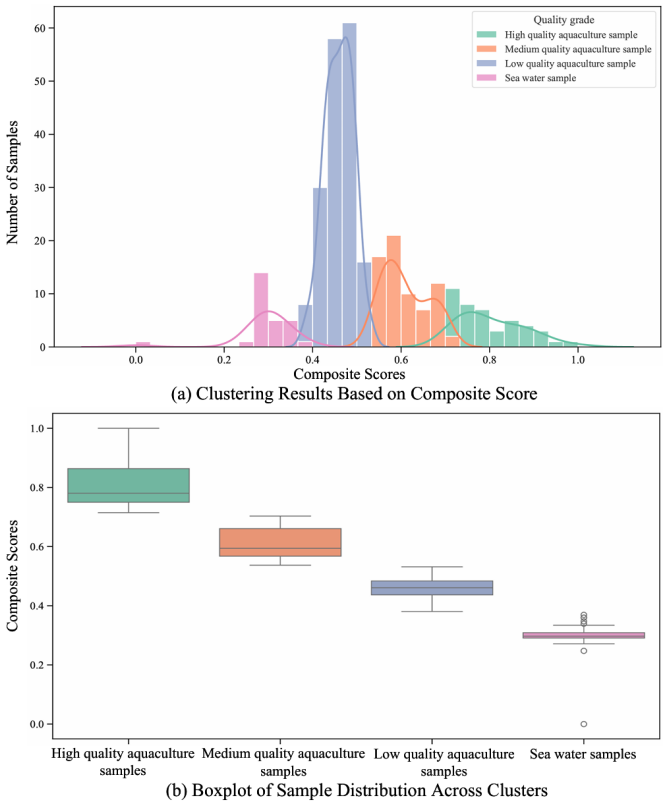


Fig. 4. Clustering results of the composite score and boxplot of sample distribution across clusters in Radarsat-2 marine SAR aquaculture dataset.

relative weights of $Mean$, MSE, C, G, IE and Gamma fitting error are 0.3221, 0.3387, 0.3273, 0.3247, -0.0164, and -0.2963, respectively. G is sensitive to SAR speckle noise and represents high-frequency spatial variation rather than pure edge quality. Since stronger speckle and clutter cause larger deviations from the reference SAR background distribution, the negative weight of the Gamma fitting error helps mitigate possible overestimation caused by high G values, together with the other metrics in the composite score. And the final values of λ_1, λ_2, and λ_3 are 0.39, 0.52, and 0.69.

In addition, clustering is also performed on the composite score metric of Radarsat-2 (RD-2) dataset to verify whether outliers exist between low-quality aquaculture samples and seawater. Both the clustering plots and boxplots are generated for analysis in Fig. 4. As shown in Figs. 3(a) and 4(a), the main difference between the Radarsat-2 and GF-3 clustering results lies in the distribution of composite scores for seawater areas: in Radarsat-2, the scores are mainly distributed between 0.2–0.4, while in GF-3 they range from 0–0.2, and after adjustment, from 0–0.39. Overall, the results are still largely consistent. This difference may be attributed to the relatively small number of seawater samples in the Radarsat-2 images, as well as the larger proportion of areas affected by wind and waves. In GF-3, bright seawater samples also exist but are more often misclassified as low-quality aquaculture samples. As shown in Figs. 3(b) and 4(b), outliers are present between seawater and low-quality samples in both satellite datasets.

Finally, the K-W test is used to verify the significance of differences in each metric among seawater, high-, medium-, and low-quality aquaculture samples, with the results shown in Table I. According to the K-W test results, the p-values of all metrics in both datasets equal 0, indicating statistically significant differences among quality tiers. The H-values are relatively large, ranging from approximately 150–240 in the Radarsat-2 dataset and from 3100–8300 in the GF-3 dataset, suggesting the five metrics possess strong discriminative power for aquaculture sample quality grading. The much smaller H-values in Radarsat-2 compared with GF-3 may be attributed to the nature of the K-W statistic, which measures rank-sum differences. As the total sample size increases, rank-sum differences are more likely to be amplified. Since the Radarsat-2 sample size is smaller than GF-3, its corresponding H-values are relatively lower.

TABLE I
KRUSKAL-WALLIS TEST RESULT

Dataset	Statistic[1]	$Mean$	MSE	C	G	IE
GF-3	H	3837	7415	8260	3132	3448
	P	0	0	0	0	0
RD-2	H	167	242	193	158	224
	P	0	0	0	0	0

[1] H is the K–W statistic measuring rank differences among groups, with larger values indicating greater inter-group separation. P denotes the probability that the observed differences are due to chance. Here, P=0 indicates that P is extremely small (much less than 0.0001), showing that the differences among quality tiers are highly statistically significant.

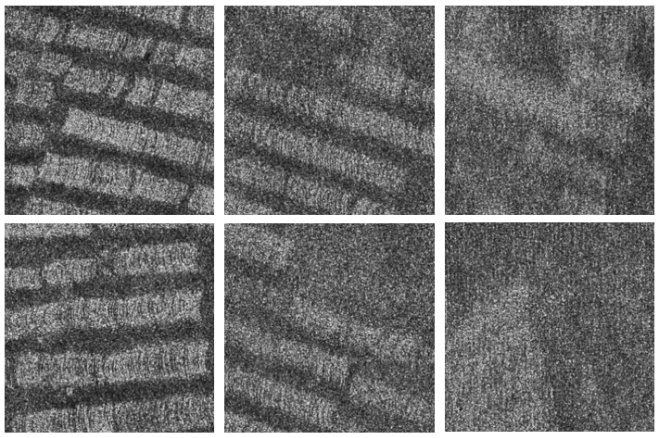
3) Automated Classification Pipeline: Due to variations between different satellite datasets, fixed thresholds may not

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TABLE II
CLASSIFICATION RESULTS ON GF-3 AND RADARSAT-2 DATASETS

Dataset	$OA(\%)$	$R(\%)$	$P(\%)$	F_1
GF-3	89.25	88.77	90.25	0.8934
RD-2	94.46	94.53	96.81	0.9399



(a) High quality samples (b) Medium quality samples (c) Low quality samples

Fig. 5. Intuitive vision of high, medium, and low-quality of marine aquaculture SAR images.

generalize well across data sources. Therefore, an automated classification pipeline is trained using the five metrics derived from GF-3 dataset. The training set comprised 9,336 images, while the test set included 2,333 GF-3 images and 300 Radarsat-2 images. The results in Table II indicate that the automated classification pipeline achieves high performance. The data quality tiers obtained from the automated classification pipeline are shown in Fig. 5. High-quality aquaculture samples exhibit high contrast, clear gray-level differences between rafts and seawater, well-defined textures, and intact raft structures, making them highly distinguishable. Medium-quality samples show generally visible raft contours, although certain regions are affected by noise. Low-quality samples display low contrast, minimal gray-level differences between rafts and seawater, blurred textures, and severe loss of structural details, making raft contours difficult or even impossible to discern.

4) Module Performance Evaluation: For the GF-3 dataset, 2,501 images selected are used for self-supervised pretraining in the upstream network, including 1,189 high-quality aquaculture images and 1,312 seawater images. For downstream fine-tuning, 373 images with ground-truth labels were used, and 93 aquaculture images are reserved for testing. As a comparison, 11,669 unselected images, including 5,775 aquaculture images and 5,894 water images, are used for upstream pretraining, while the downstream fine-tuning and test sets remained the same. For the Radarsat-2 dataset, 986 images selected are used for upstream self-supervised training, including 493 high-quality aquaculture images and 493 seawater images. Downstream fine-tuning used 200 images with ground-truth labels, and 93 floating raft aquaculture images are used for testing. To validate the superiority of the proposed QADSM, we evaluate it against the uncurated dataset and several representative data selection baselines, including random selection, SAR-domain quality-based resampling [42], and three general-purpose coreset selection methods [38]–[40]. The comparative results are presented in Table III, where QADSM consistently achieves the best performance across both datasets while using only 20%–30% of the total pretraining data. Two observations further highlight the advantage of QADSM. First, the general-purpose coreset methods remain consistently inferior to QADSM across both datasets. This suggests that sample selection strategies designed for generic visual representations are not necessarily optimal for marine SAR imagery. Second, even the SAR-oriented entropy-based method underperforms QADSM by 2.34% mIoU on GF-3 and 2.84% mIoU on Radarsat-2, as a single-metric quality proxy cannot fully identify the most informative SAR samples for SSL pretraining. These results confirm that QADSM's domain-adaptive, multidimensional quality assessment is essential for efficient and effective SSL pretraining in marine SAR imagery.

TABLE III
PERFORMANCE EVALUATION OF THE QUALITY-AWARE DATA SELECTION MODULE ON GF-3 AND RADARSAT-2 DATASETS.

Dataset	Method	$mIoU$	$Kappa$	$OA(\%)$	$R(\%)$	$P(\%)$	F_1
GF-3	Uncurated	0.7405	0.6976	85.47	88.17	85.47	0.8646
	Random	0.7212	0.6752	83.80	86.19	79.54	0.8273
	Entropy-based [42]	0.7274	0.6823	84.22	84.33	81.32	0.8279
	Moderate Coreset [38]	0.7227	0.6767	83.90	85.32	80.19	0.8268
	CCS [39]	0.7230	0.6768	83.92	84.86	80.48	0.8261
	ELFS [40]	0.7277	0.6828	84.24	84.60	81.18	0.8286
	QADSM (ours)	**0.7508**	**0.7113**	**86.32**	**89.38**	**86.07**	**0.8743**
RD-2	Uncurated	0.7619	0.6949	86.49	**96.45**	83.51	0.8951
	Random	0.7698	0.6875	86.99	86.20	89.83	0.8798
	Entropy-based [42]	0.7700	0.6858	87.01	85.04	90.63	0.8775
	Moderate Coreset [38]	0.7691	0.6910	86.95	88.64	88.28	0.8846
	CCS [39]	0.7707	0.6902	87.05	87.00	89.50	0.8823
	ELFS [40]	0.7685	0.6823	86.91	84.26	90.87	0.8744
	QADSM (ours)	**0.7984**	**0.7286**	**88.79**	87.78	**93.95**	**0.9076**

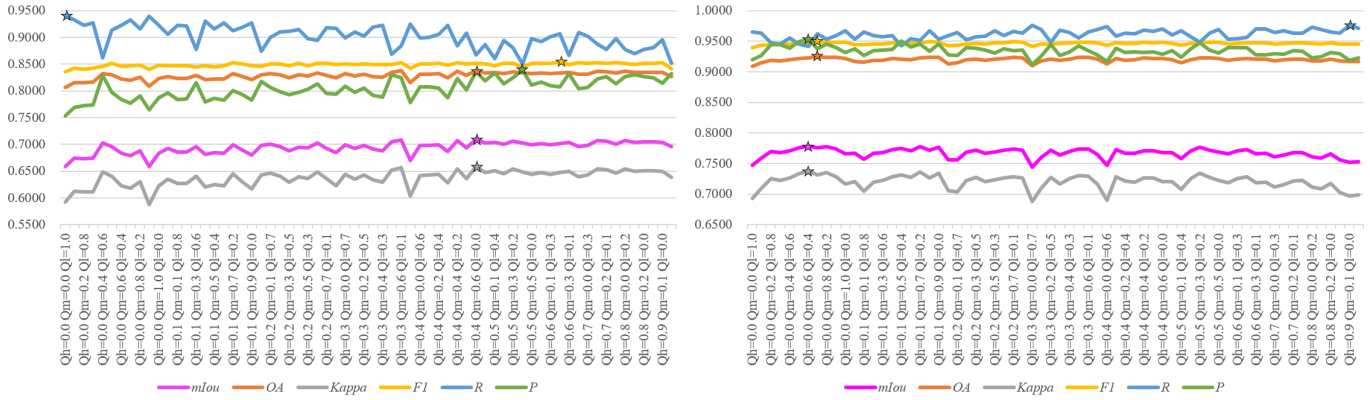
C. Principled Data Proportioning

1) Results of Data Proportioning Experiments: Using the pre-trained weights obtained from the QADSM module, data proportioning experiments are conducted on both the GF-3 and Radarsat-2 datasets. For GF-3, 373 images with different proportions of high, medium, and low-quality samples are randomly selected for fine-tuning. The trained models are then evaluated on a mixed-quality dataset (31 images from each quality tier). For Radarsat-2, 200 images with different proportions of high, medium, and low-quality samples are randomly selected for fine-tuning, and the models are also evaluated on a mixed-quality dataset (31 images from each quality tier). The results are shown in Fig. 6. Furthermore, experiments are carried out on datasets of different quality tiers, including high-quality (31 images), medium-quality (31 images), and low-quality (31 images). The detailed results can be found in the supplementary material.

From Fig. 6, several key observations can be made:

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(a) GF-3 (b) Radarsat-2

Fig. 6. GF-3 and Radarsat-2 Dataset Proportioning Results.

- Sample proportions have a profound impact on model performance. For the GF-3 dataset, $mIoU$ ranges from 0.6585 to 0.7093, OA from 80.84% to 83.81%, $Kappa$ from 0.5871 to 0.6581, and F_1 from 0.8361 to 0.8536. For the Radarsat-2 dataset, $mIoU$ ranges from 0.7440 to 0.7786, OA from 90.98% to 92.25%, $Kappa$ from 0.6881 to 0.7378, and F_1 from 0.9397 to 0.9494.
- For the 16 lowest-performing sample proportion configurations, the GF-3 dataset is characterized by a low proportion of high-quality samples (≤ 0.2), whereas the Radarsat-2 dataset shows a low proportion of medium-quality samples (< 0.2). In contrast, for the 16 highest-performing configurations, both datasets exhibit low-quality sample proportions not exceeding 0.5.
- When the proportion of high-quality samples is zero, increasing the proportion of medium-quality samples leads to a "rise-then-fall" trend in overall performance.
- When the proportion of high-quality samples exceeds 0.4, fluctuations in model performance are noticeably reduced, and results become more stable.
- In the Radarsat-2 dataset, model performance exhibits periodic variations with changing sample proportions. When the proportion of high-quality samples is fixed and the medium-quality proportion is near zero, model accuracy reaches its minimum.

2) Contribution Analysis of Quality Tiers: By analyzing the experimental results of GF-3 and Radarsat-2 under different quality sample proportions, a marginal effect pattern in the data quality distribution is observed, manifested as diminishing marginal returns for high, medium, and low-quality samples.

When the proportion of high-quality samples is already high, further increasing their ratio yields only marginal improvement in model performance. This reflects the diminishing marginal contribution of high-quality data: once the model has sufficiently learned the features of high-quality samples, the information gain from additional samples of the same type becomes limited. This phenomenon is particularly pronounced in the GF-3 dataset.

In scenarios with insufficient high-quality samples, medium-quality samples can effectively compensate for the diversity of the training data, significantly enhancing model performance.

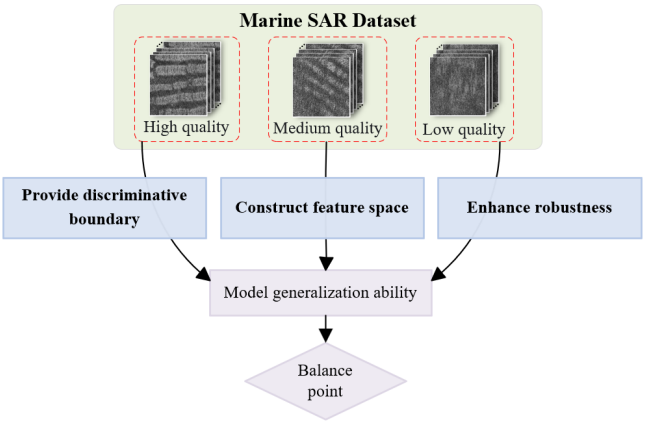
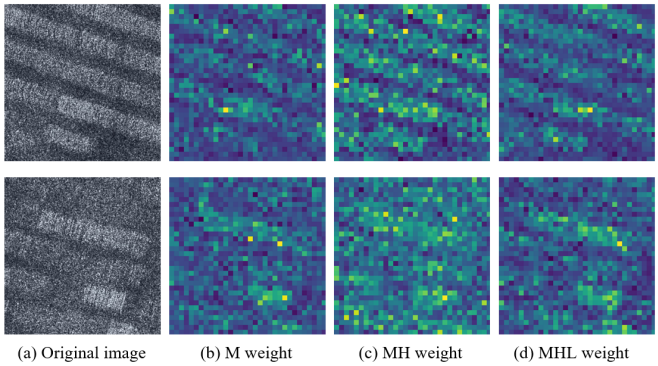


Fig. 7. Contribution of Quality Tiers.



(a) Original image (b) M weight (c) MH weight (d) MHL weight

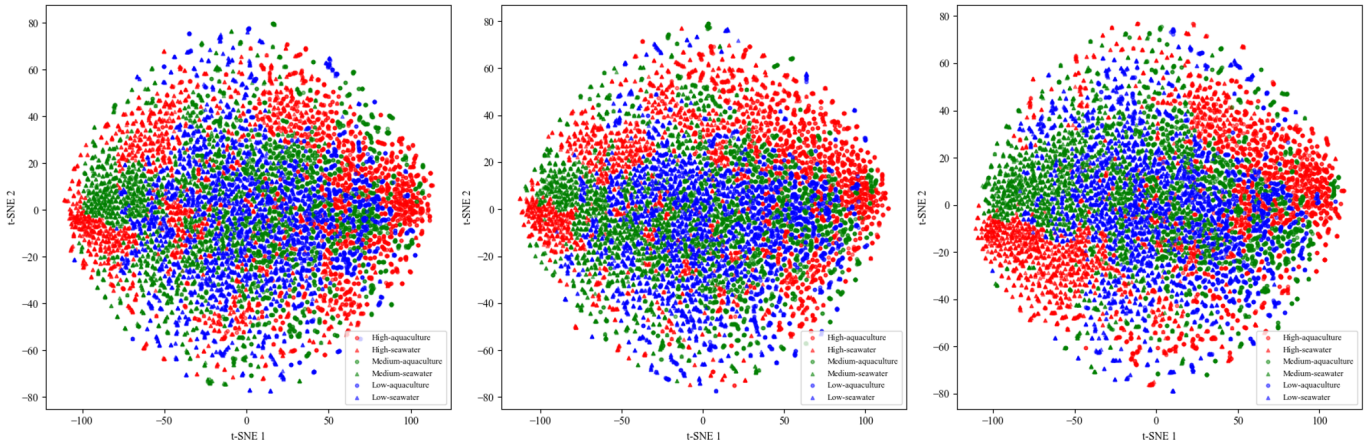
Fig. 8. The visualization results of the three sets of weights obtained during the training process.

However, when the proportion of medium-quality samples exceeds a certain threshold, accumulated noise begins to negatively impact model training, leading to a decline in performance, resulting in a "gain-then-loss" curve.

As the proportion of low-quality samples increases, it can initially help increase sample diversity, and its negative effects may not be obvious at first. However, once their proportion exceeds 50%, the model's discriminative ability is significantly weakened, and performance deteriorates rapidly. This

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(a) M weight (b) MH weight (c) MHL weight

Fig. 9. T-SNE visualization of pixel-level feature spaces for four samples each from high, medium, and low-quality data under different weights.

is because low-quality data often contain substantial noise and labeling errors, whose cumulative effect amplifies model errors, producing a negative marginal effect.

Based on these observations, the following hypothesis is proposed: during model training, **medium-quality data mainly contribute to constructing the feature space, high-quality data provide discriminative boundaries, and low-quality data enhance robustness**, as shown in Fig. 7. Accordingly, the downstream network is first trained on medium-quality data, then further trained with the addition of high-quality data, and finally with low-quality data. This sequential training produces three sets of weights, denoted as M, MH, and MHL, respectively. The weights of the three are visualized to verify the hypothesis, as shown in Fig. 8. Across all the visualization results, it can be seen that after training with medium-quality data, the model is already able to roughly distinguish between seawater and aquaculture areas in Fig. 8(b). After adding high-quality data, the edges in the images become sharper in Fig. 8(c). Incorporating low-quality data further filters out many regions that were mistakenly identified as aquaculture areas in Fig. 8(c), as shown in Fig. 8(d).

Furthermore, the pixel-level feature spaces of high, medium, and low-quality data are visualized under different weighting (M, MH, MHL), as illustrated in Fig. 9. And the image-level feature space of high, medium, and low-quality data are under the MHL weight, as shown in Fig. 10. Data of different quality tiers contribute differently in the feature space, as shown in Fig. 9. Medium-quality data (green) form a uniform distribution, establishing the main spatial framework. High-quality data (red) cluster tightly, creating clear inter-class boundaries, while low-quality data (blue) spread widely, enhancing diversity and introducing noise. Notably, in Fig. 9(c), the training strategy combining high, medium, and low-quality data produces more compact high-quality clusters with sharper boundaries, indicating that multi-quality collaborative training extracts more discriminative features. The dimensionality-reduced visualization in Fig. 10 confirms this pattern: high-quality data form tight clusters, medium-quality data shape the framework, and low-quality data expand

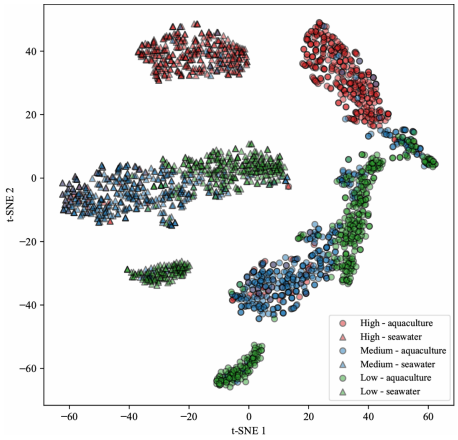


Fig. 10. T-SNE visualization of image-level feature space for 373 samples each from high, medium, and low-quality data under the MHL weight.

feature coverage, supporting the mechanism of "high-quality defines boundaries, medium-quality builds the framework, and low-quality enhances robustness."

Since t-SNE emphasizes local structure, cluster shapes and distances do not strictly reflect high-dimensional relationships. Thus, the visualizations only indicate general trends and require further quantitative verification. The clustering performance of different quality data under the MHL weights is evaluated using the silhouette coefficient (SC), centroid distance (CD), and intra-class variance (IV), as shown in Table IV. Meanwhile, the robustness under different training weights is evaluated by σ, as reported in Table V. Specifically, the silhouette coefficient assesses intra-cluster cohesion and inter-cluster separation, centroid distance quantifies separation between classes or clusters, intra-class variance evaluates within-class tightness and consistency, and robustness (σ) measures output stability under perturbations.

As shown in Table IV, data of different quality tiers exhibit distinct distribution patterns and functional roles in model learning. High-quality data (silhouette coefficient 0.69, intra-class variance of aquaculture 2.70) form compact clusters with well-defined boundaries in the feature space, located at the

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TABLE IV
CLUSTERING METRICS OF HIGH, MEDIUM, AND LOW-QUALITY DATA UNDER MHL WEIGHT.

Quality	SC	CD	IV (aquaculture)	IV (seawater)
High	**0.6913**	**6.1485**	**2.7041**	1.5495
Medium	0.6080	5.7085	6.1427	**1.4836**
Low	0.4373	5.0518	10.6485	1.9788

TABLE V
ROBUSTNESS $\sigma_{0.05}$ OF HIGH, MEDIUM, AND LOW-QUALITY DATA UNDER DIFFERENT WEIGHTS.

Weight	Robustness_σ_0.05		
	High	Medium	Low
M	0.01546	0.01707	0.01698
MH	0.01532	0.01744	0.01683
MHL	**0.01483**	**0.01661**	**0.01679**

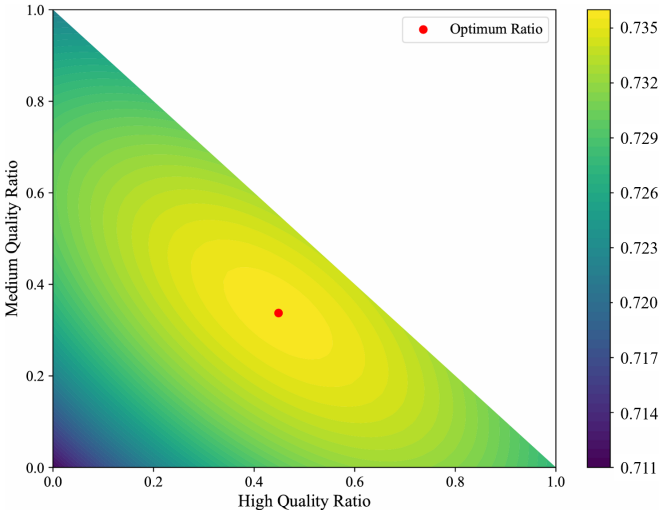


Fig. 11. Heatmap of model performance ($mIoU$) under different proportion.

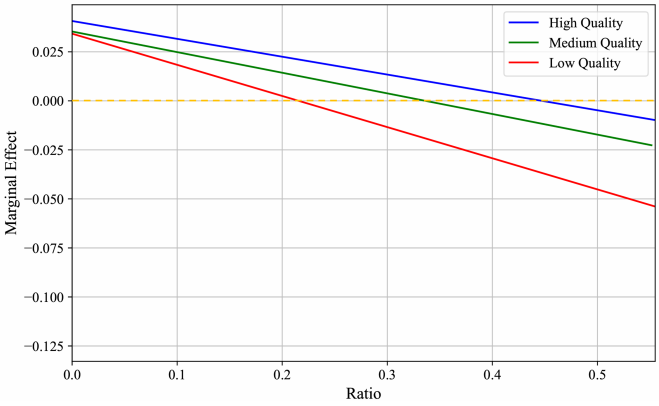


Fig. 12. Marginal effects across data quality tiers.

outer region (centroid distance 6.15). Their highly concentrated distribution provides precise discriminative boundaries for the model. Medium-quality data (intra-class variance of aquaculture 6.14) constitute the core body of the feature space, characterized by a moderate centroid distance (5.71) and broad coverage, serving as the fundamental component of the feature space structure. Low-quality data (silhouette coefficient 0.437, intra-class variance of aquaculture 10.65, centroid distance 5.05) present a "dense center with diffuse periphery" pattern, and their extensive coverage enhances model robustness. As indicated in Table V, under a noise level of $\sigma = 0.05$, models trained with a combination of all three quality tiers exhibit stronger resistance to interference compared with those trained solely on medium-quality or medium–high-quality data. Collectively, the three quality tiers establish a hierarchical structure in the feature space: low-quality data form an outer noise halo, high-quality data define the boundary layer, and medium-quality data shape the central clusters. This synergistic mechanism substantiates the theoretical hypothesis of "medium-quality data constructing the feature space, high-quality data providing discriminative boundaries, and low-quality data enhancing robustness", thereby offering theoretical support for optimizing data quality proportioning.

3) Recommended Proportion: The experimental results indicate that different performance evaluation metrics exhibit strong consistency in their overall trends. $mIoU$ stands out as one of the most important and widely adopted metrics, as it provides a comprehensive measure of the overlap between predictions and ground truth. Accordingly, $mIoU$ is chosen as the core evaluation metric, and a parameterized fitting model is constructed to capture its relationship with data quality proportions. The fitted parameter vector was obtained as $\theta = [0.0615, -0.0455, 0.0630, -0.0526, -0.0617, 0.7114]^\top$. The optimal data-quality proportions found are: high-quality 45%, medium-quality 34%, and low-quality 21%. Note that the optimal proportions may vary across datasets.

Based on these results, the $mIoU$ heatmap over data-quality proportions and the marginal-effect curves for each quality tier are plotted in Fig. 11 and Fig. 12 to illustrate performance variations with data mix. Fig. 11 shows a clear nonlinear dependency of $mIoU$ on high and medium-quality sample ratios: $mIoU$ is low when the low-quality proportion is large, and reaches its maximum in regions with moderate shares of high and medium-quality samples. From the contour density, $mIoU$ is more sensitive to changes in medium-quality proportion when the high-quality share is low; when the high-quality share is sufficiently large, $mIoU$ becomes less sensitive to medium-quality fluctuations, indicating that a certain fraction of high-quality samples stabilizes performance. Fig. 12 shows that the marginal effects of high, medium, and low-quality samples all decline as their proportions increase. The marginal benefit of high-quality samples is largest at low proportions, yielding the most pronounced $mIoU$ gains when added in small amounts. Medium-quality samples exhibit a steeper decline in marginal effect than high-quality ones: a small amount of medium-quality data can substantially affect performance, but gains quickly saturate. Low-quality samples show the steepest decline in marginal effect, implying that excessive low-quality data degrades model performance, although small proportions help expand feature coverage and improve robustness. The optimal proportion corresponds to points where the marginal gains of the three sample types

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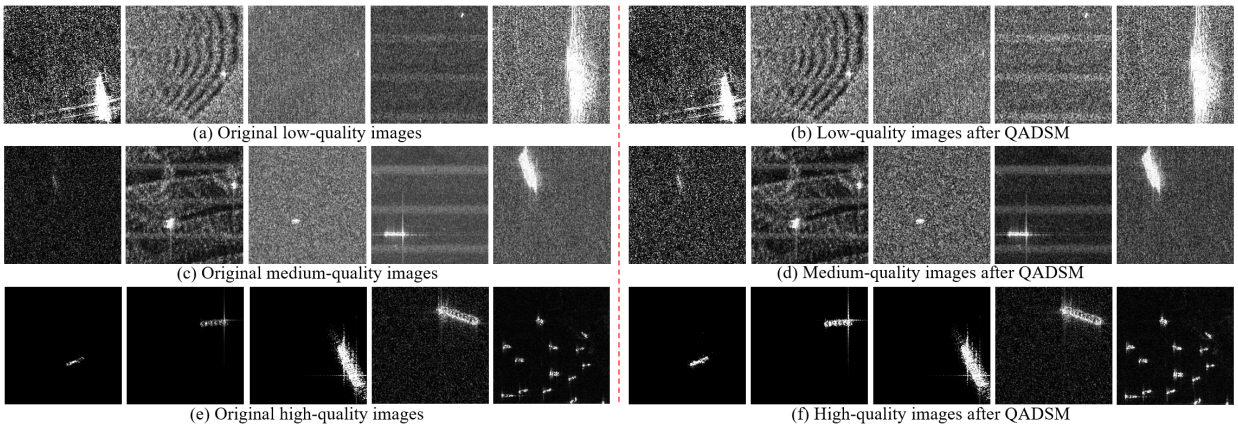


Fig. 13. Intuitive vision of high, medium, and low-quality of MSAR ship subset images.

approach zero, indicating a balance of marginal benefits and a peak in model performance. Overall, the data proportion substantially influences training outcomes; maintaining appropriate proportions of high, medium, and low-quality samples is crucial for both accuracy and stability.

To further validate the practical usefulness of the recommended proportioning strategy, we compare three fine-tuning strategies: using all available annotated samples, using only high-quality samples, and using samples allocated according to the recommended quality proportion. As shown in Table VI, the recommended proportion consistently achieves the best overall performance on both GF-3 and Radarsat-2 datasets. These results indicate that simply increasing the amount of annotated data or selecting only high-quality samples is not necessarily optimal. Although high-quality samples provide clean and discriminative information, relying exclusively on them limits distributional diversity. In contrast, an appropriate mixture of high, medium, and low-quality samples introduces complementary information, leading to better generalization under a limited annotation budget.

TABLE VI
COMPARISON OF FINE-TUNING STRATEGIES ON GF-3 AND RADARSAT-2 DATASETS.

Dataset	Strategy	$mIoU$	$Kappa$	$OA(\%)$	$R(\%)$	$P(\%)$	F_1
GF-3	All annotated	0.6889	0.6300	82.62	**91.92**	78.58	0.8496
	High-only	0.6958	0.6381	82.68	84.16	**85.14**	0.8324
	Recommended	**0.7066**	**0.6544**	**83.70**	89.23	81.26	**0.8539**
RD-2	All annotated	0.7688	0.7218	92.12	95.75	93.83	0.9469
	High-only	0.7530	0.6984	91.67	**97.15**	92.26	0.9453
	Recommended	**0.7752**	**0.7321**	**92.26**	95.53	**94.19**	**0.9478**

D. Validation on the MSAR Object Detection Dataset

Beyond the GF-3 and Radarsat-2 semantic segmentation datasets, the processed MSAR ship subset provides a SAR object detection benchmark for evaluating the generalization ability of QuaSAR. QADSM is first applied to stratify the MSAR ship subset into high, medium, and low-quality tiers. As shown in Fig. 13, representative samples from the three tiers exhibit clear visual differences in target-background contrast, texture clarity, and structural completeness. Detailed quality-division process and results, including clustering distributions and K-W tests, are provided in the supplementary material.

For the upstream data selection experiment, 5,532 selected high-quality ship images are used for self-supervised pretraining, while 14,660 unfiltered ship images are used as the uncurated baseline. For a fair comparison, both settings use the same 1,293 randomly selected annotated images for downstream fine-tuning and the same 324 randomly selected images for testing. The quantitative results of the object detection task are summarized in Table VII. Compared with the uncurated baseline, QADSM significantly improves high-precision localization, with AP_{75} increasing from 13.0 to 16.2, and enhances small-object detection, with AP_S improving from 11.4 to 14.0. These gains are attributed to the cleaner structure and reduced speckle interference of the selected high-quality samples, which facilitate more accurate feature learning. A slight drop in AP_{50} and AP_L is observed, which stems from the reduced pretraining data size (only about 37.7% of the data). While the uncurated dataset provides greater diversity that benefits large-object and loose IoU performance, QADSM achieves a favorable trade-off by filtering redundant samples and improving fine-grained detection accuracy with substantially higher data efficiency.

TABLE VII
PERFORMANCE EVALUATION OF THE QUALITY-AWARE DATA SELECTION MODULE ON THE MSAR DATASET.

Dataset	Method	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
MSAR	Uncurated	**52.2**	13.0	11.4	45.8	**69.4**
	QADSM	49.3	**16.2**	**14.0**	**46.1**	61.4

Following the upstream pretraining phase, data proportioning experiments are conducted on the MSAR dataset using the pre-trained weights obtained from the QADSM module. For downstream fine-tuning, 1,293 images with different proportions of high, medium, and low-quality samples are randomly selected, and the trained models are evaluated on a mixed-quality test set containing 324 images, with 108 images from each quality tier. AP_{50} is adopted as the core metric to fit the relationship between detection performance and data-quality

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proportions, yielding an optimal proportion of approximately 30% high-quality, 42% medium-quality, and 28% low-quality samples. To verify its effectiveness, three fine-tuning strategies are compared: using all annotated samples, using only high-quality samples, and using samples allocated according to the recommended proportion. The results are summarized in Table VIII, while the complete proportioning results and marginal-effect analysis are provided in the supplementary material. As shown in Table VIII, the recommended proportion achieves the best AP_{50}, AP_S, and AP_L, whereas the high-only strategy performs worst across all metrics. This trend is consistent with the findings on the GF-3 and Radarsat-2 datasets, suggesting that clean samples alone cannot fully cover the scale and appearance variations of ships in MSAR. Although the all-annotated strategy shows slight advantages in AP_{75} and AP_M, its lower AP_{50} indicates that simply increasing annotated data may introduce redundant or less informative samples. These results further validate that PDPS can generalize to SAR object detection by balancing data quality, target-scale coverage, and detection accuracy.

TABLE VIII
COMPARISON OF FINE-TUNING STRATEGIES ON THE MSAR DATASET.

Dataset	Strategy	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
MSAR	All annotated	49.2	**13.7**	12.1	**47.0**	48.4
	High-only	38.8	11.9	10.5	38.1	31.5
	Recommended	**52.6**	12.2	**14.5**	43.3	**48.9**

V. CONCLUSION

This paper presents QuaSAR, a quality-aware selection and proportion framework for SSL in marine SAR imagery. The main conclusions are as follows:

1) **Quality-aware data selection:** A quality-aware selection mechanism is designed to stratify marine SAR images, effectively identifying high-quality samples, which accelerates and stabilizes the SSL pre-training process.
2) **Insights into data contributions:** A principled proportioning strategy is proposed, and the contributions of high, medium, and low-quality data are analyzed, whereby their distinct roles are revealed: high-quality data are found to define decision boundaries, medium-quality data to construct the feature space, and low-quality data to enhance robustness.
3) **Recommended data proportioning:** A practical data proportioning algorithm is provided based on these insights, guiding the efficient use of limited SAR data resources.
4) **Experimental validation:** Extensive experiments on the GF-3, Radarsat-2, and MSAR datasets are conducted, which demonstrated that QuaSAR not only improves training efficiency and model performance but also offers practical guidance for the optimal proportion of limited data.

In summary, QuaSAR provides a scalable and data-driven solution for SSL in marine SAR imagery, and emphasizes the critical role of data selection and proportioning in enhancing generalization, stability, and robustness.

Despite these advantages, a limitation of the QuaSAR framework is that it is tailored to marine SAR imagery with relatively clear target-background backscattering contrast. It is not directly applicable to land SAR scenes, where heterogeneous backgrounds and complex scattering mechanisms may affect quality evaluation. In future work, we will explore scene-adaptive quality metrics to extend QuaSAR to more complex SAR scenarios.

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Yizhi Zhao is currently working toward the PhD degree with the School of Control Science and Engineering, Dalian University of Technology. She received the M.S. degree in physical oceanography from the Second Institute of Oceanography, Ministry of Natural Resources, Hangzhou, China, in 2023. Her research interests include self-supervised deep learning and remote sensing image processing.



Jianchao Fan (S'10–M'12–SM'25) received the B.S. in automatical technology and Ph.D. degrees in control theory and control engineering from the School of Control Science and Engineering, Dalian University of Technology, Dalian, China, in 2007 and 2012, respectively. He was a Post-Doctoral Fellow jointly cultivated by The Chinese University of Hong Kong and Dalian University of Technology from 2013 to 2016, and also a Visiting Scholar with the Department of Computer Science and Engineering, Washington University in St. Louis, St. Louis, MO, USA, from 2018 to 2019. He is currently a Professor with School of Control Science and Engineering, Dalian University of Technology. He has been a Professor with the Department of Ocean Remote Sensing, National Marine Environmental Monitoring Center, Dalian, China, since 2019. He has authored/co-authored 4 books and over 130 scientific papers in international journals and conference proceedings, such as IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON FUZZY SYSTEMS, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, *etc*. His current research interests include marine SAR image processing, polarimetric SAR, deep learning neural networks, and heuristic swarm intelligent optimization.

Dr. Fan serves as Program Chair, Workshop Chair, the Publicity Chair, and the Local Arrangement Chair of several international conferences, such as the International Symposium on Neural Networks, the International Conference on Information Society and Technology, and the International Conference on Intelligent Control and Information Processing.

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